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课题研究报告 · RESEARCH STUDY

AI增长工程：从漏斗到闭环

AI Growth Engineering: From the Funnel to the Loop

关于发现线中介化、代理用户与结果留存的课题研究

A research study on the mediated discovery line, the agent as user, and retention as outcome

研究底本 SOURCE
Cuofano, G. AI Growth Engineering (2026)

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摘要 *Abstract*

本研究考察答案引擎与AI代理站到企业与买家之间后，增长的对象如何改变。研究以对Cuofano《AI Growth Engineering》（2026）的结构化分析性阅读为方法，结合底本报告的合成重建与截至2026年第三季度的公开案例，提出四项发现。**第一**，发现线被中介化，“被引用”取代“被点击”成为可见性的单位，企业须逐项资产划定产权线。**第二**，越来越多的用户是阅读规格的代理，产品需要第二表面，代理同期群须单独计量，定价成为闭环选择器。**第三**，留存以持续交付的结果衡量，激活即首个经验证结果，残差闭环是唯一无法被机器中介的复利来源。**第四**，以人类会话为前提的指标已与价值脱钩，智能体速度的实验需要冻结的反事实裁判。案例显示，AI原生产品的增长来自代理表面与模型发布，并因免费用户成本化而转向企业。研究最后讨论按闭环组织团队的启示、研究局限与未来议程。

This study examines how the object of growth changes once answer engines and AI agents stand between firms and their buyers. Through a structured analytical reading of Cuofano's *AI Growth Engineering* (2026), with the composite rebuild and dated public cases it reports as of Q3 2026, it advances four findings. **First**, discovery is mediated: being cited has replaced being clicked as the unit of visibility, and firms must draw a property line asset by asset. **Second**, a growing share of users are agents that read specifications; products need a second surface, the agent cohort must be counted separately, and pricing becomes a loop selector. **Third**, retention is measured as outcomes kept; activation is the first verified outcome, and the residue loop is the one compounding source machines cannot mediate. **Fourth**, session-premised metrics have decoupled from value, and agent-speed experiments need a frozen counterfactual referee. The cases suggest AI-native products grow through agent surfaces and model releases and move toward enterprise because the free user has become a cost. The study closes with implications for organising by loop, limitations and a research agenda.

关键词 增长工程；发现线；代理即用户；结果留存；残差闭环；增长古德哈特

KEYWORDS growth engineering; discovery line; agent as user; outcome retention; residue loop; growth Goodhart

研究问题 *Research Questions*

RQ1	当答案引擎中介了发现线，企业如何获取并衡量可见性?	When answer engines mediate discovery, how do firms win and measure visibility?
RQ2	当用户是代理而非人，产品表面、报价与定价应如何重新设计?	When the user is an agent rather than a person, how should product surfaces, offers and pricing be redesigned?
RQ3	以结果衡量的留存如何改变激活定义与增长的单位经济?	How does outcome-based retention change the definition of activation and the unit economics of growth?
RQ4	在智能体速度的优化压力下，增长度量如何保持有效?	How can growth measurement stay valid under optimisation pressure at agent speed?

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第 1 章 · CHAPTER 1

01 研究背景与问题提出

Background and Problem Statement

增长职能的仪表盘仍在“正常”读数，而它所测量的对象已经移动。本章追溯这一错位的来源，并提出本研究的问题。

Growth dashboards kept reading normally while the object they measured moved. This chapter traces the source of that mismatch and states the questions of the study.

中文

ENGLISH

1.1 一个仍在“正常”读数的学科

A Discipline Whose Instruments Still Read Normally

底本认为，在企业各职能中，增长是最后一个意识到AI时代改写了自身“物理规律”的部门。流量照常到达，注册照常发生，参与度曲线仍在上扬，仪表盘上没有任何信号提示断裂。然而在表层之下，三项倒置已经改写了“增长”所指涉的对象：谁站在发现的入口，谁是用户，以及何为留存。一个学科若在指涉对象移动时仍维持“正常读数”，便会陷入一种特定的失败：它以越来越高的效率，去优化那些已不再反映价值的数字。

In the source's account, growth was the last function of the firm to notice that the AI era had altered its physics. Traffic still arrived, sign-ups still registered and engagement curves still climbed, so nothing on the dashboard signalled a break. Underneath, three inversions had rewritten what growth refers to: who stands at the point of discovery, who the user is, and what counts as retention. A discipline whose instruments keep reading normally while their referent shifts is exposed to a specific failure: it optimises numbers that no longer track value, and does so with growing efficiency.

1.2 Web平方：由外而内与由内而外

Web Squared: Outside-In and Inside-Out

底本将这一转变置于一个更宏观的命题之下，称之为“Web平方”（Web Squared）：AI并未取代互联网，而是在其之上复利增长。当下的模型之所以存在，是因为互联网生产并分发了训练语料；AI接入互联网既有的分发基础设施后，产生的增长曲线往往比互联网本身更陡峭。因此，率先被颠覆与放大的正是互联网所催生的行业——搜索、社交、出版、云与软件——它们早已掌握智能可以直接放大的数据、分发与经济结构。

The volume places this shift inside a broader thesis it calls Web Squared: AI does not replace the web, it compounds it. Today's models exist because the web produced and distributed their corpus, and AI, plugged into the web's distribution infrastructure, generates growth curves steeper than the web itself usually did. The industries the web created—search, social, publishing, cloud and software—are therefore the first to be disrupted and amplified, because they already hold the data, distribution and economics that intelligence can immediately multiply.

对增长而言，第二个区分更为关键。互联网是一种“由外而内”的技术：最先改变的是分发，而组织的运营模式在此后多年里基本维持原状，直到2010年代才真正触及商业模式的核心。AI则是“由内而外”的：嵌入组织之后，它首先改变工作方式与决策方式，继而重塑商业模式，最后才重新定义分发。底本由此推出的含义是，这一时代的增

For growth, the second distinction matters more. The web was an outside-in technology: distribution changed first, while operating models stayed largely intact for years, reaching the core of business models only in the 2010s. AI works inside-out. Embedded in an organisation, it first changes how work is done and decisions are made, then reshapes the business model, and only last redefines distribution. The implication the source draws is that the growth loop now starts in

长闭环始于运营模式，而非企业的边缘；这也解释了为何企业级AI落地艰难——组织转型是商业模式转型的前提。

the operating model rather than at the edge of the firm—which also explains why enterprise AI is hard: organisational transformation is the prerequisite for business-model change.

表 1-1 两种技术扩散方向的比较
Table 1-1 *Two directions of technological diffusion*

维度 Dimension	互联网：由外而内 Web: outside-in	AI：由内而外 AI: inside-out
最先改变 Changes first	分发渠道 Distribution	运营模式：工作与决策 Operating model: work and decisions
随后改变 Changes next	多年后才触及商业模式 Business model, years later	价值创造与获取方式 How value is created and captured
最后改变 Changes last	组织的运营核心 The operating core	分发与发现线 Distribution and the discovery line
增长学科的核心课题 Core task of growth	让产品与分发一同复利 Make product and distribution compound together	将智能、产品、运营、商业模式与分发连成一个复利系统 Join intelligence, product, operations, business model and distribution into one compounding system

来源 Source: Cuofano (2026), Prologue；本研究整理 tabulated by this study

1.3 三项倒置
Three Inversions

- **发现线的中介化**：答案引擎与助手站在企业与买家之间，“被引用”取代“被点击”成为可见性的基本单位。
- **代理即用户**：越来越多到达产品的不是人，而是人的代理——它阅读定价页、调用接口、比较报价并完成购买。
- **留存即结果**：当软件替人完成工作，登录几乎说明不了什么；留存取决于结果是否持续到来。

- **The mediated discovery line**: answer engines and assistants stand between firm and buyer; being cited replaces being clicked as the unit of visibility.
- **The agent as user**: a growing share of arrivals are not people but their agents, which read the pricing page, call endpoints, compare offers and complete purchases.
- **Retention as outcome**: when software does the work, a login says little; retention is whether the outcome keeps arriving.

1.4 问题的提出
Statement of the Problem

如果点击、人类会话与回访不再是增长的锚点，那么增长职能应当构建、测量与守护的对象由什么替代？本研究围绕摘要页所列四个问题展开：发现线中介化如何改变可见性的获取；代理用户如何改变产品与报价的设计；以结果衡量的留存如何改变激活与单位经济；以及在智能体速度的优化压力下，度量体系如何保持有效。

If the click, the human session and the return visit no longer anchor growth, what replaces them as the objects a growth function builds, measures and defends? The study pursues four questions, listed on the abstract page: how mediated discovery changes the way visibility is won; how the agent-as-user changes product and offer design; how outcome-based retention changes activation and unit economics; and how measurement stays valid under optimisation pressure at agent speed.

第 2 章 · CHAPTER 2

02 核心概念与理论框架

Core Concepts and Analytical Framework

本章区分增长的“永恒物理”与AI时代的四项改写，并据此建立以五个闭环为核心的分析框架。

This chapter separates the permanent physics of growth from the four rewrites of the AI era, and builds an analytical framework around five loops.

中文　　ENGLISH

2.1 五个时代及其遗产

Five Eras and Their Legacies

底本将增长史解读为四次重建，每一次都留下了当下仍在使用的工具。直复营销与客户关系管理留下了同期群与客户生命周期价值；搜索时代持续约十五年，使“为机器读者组织好的企业实质内容”第一次成为增长引擎；社交与病毒时代把分发迁入信息流，“增长黑客”一词出现于2010年，高频实验方法在2017年被系统化；2010年代后期的闭环时代则主张漏斗是错误的形状，产品本身才是引擎，网络冷启动问题于2021年被形式化。

The source reads the history of growth as four rebuilds, each leaving an instrument the present still uses. Direct response and CRM left the cohort and lifetime value. Search, for some fifteen years, made the firm's substance—organised for a machine reader—the growth engine for the first time. The social and viral era moved distribution into feeds; "growth hacking" was coined in 2010 and high-tempo experimentation codified in 2017. The loop era of the late 2010s argued that funnels were the wrong shape and the product itself is the engine, with the cold-start problem formalised in 2021.

图 2-1 增长的五个时代：工具延续，对象改变
Figure 2-1 *Five eras of growth: instruments persist, the object changes*



来源 Source: Cuofano (2026), Ch. I

2.2 增长的永恒物理

The Permanent Physics of Growth

分析框架建立在时代未曾改变的部分之上。增长经历五个阶段——获客、激活、留存、推荐与收入——它们只在静态图中呈现为漏斗；一旦运动起来便是闭环，产品的产出（内容、数据、网络）成为下一轮的输入。四条定律支配这一闭环：留存是天花板；冷启动需要一个在小规模下即可运转的原子单元；单位经济必须闭合，回本周期决定增长是靠复利还是靠购买；任何优化过程都会博弈其度量指标。

The framework rests on what the era did not change. Growth passes through five stages—acquisition, activation, retention, referral and revenue—which form a funnel only in a static picture; in motion they are a loop in which the product's output (content, data, network) becomes the next cycle's input. Four laws govern it: retention is the ceiling; the cold start needs an atomic unit that works at small scale; unit economics must close, with payback setting whether

growth is compounded or bought; and every optimisation process games its measure.

2.3 AI时代的四项改写
Four Rewrites of the AI Era

- **发现线中介化**：答案引擎读取企业的实质内容并直接给出合成答案，承载一切增长指标的“访问”不再发生。
- **代理即用户**：机器可读的报价、可调用的接口与为代理购买设计的结算通道，正以开放协议的形式到来。
- **指标坍缩**：席位、会话与月活都假设屏幕前有一个人，而代理在两个方向上同时打破了这一假设。
- **内容商品化**：当人人都能生成无限的“看似合理”的文本，泛化内容的回报转为负值。

- **Mediated discovery**: an answer engine reads the firm's substance and delivers a synthesised answer, so the visit that carried every growth metric never happens.
- **The agent as user**: machine-readable offers, callable endpoints and settlement rails for agent purchases are arriving as open protocols.
- **Metric collapse**: seats, sessions and monthly actives assume a human, and agents break the assumption in both directions.
- **Content commoditisation**: when anyone can generate unlimited plausible text, generic content has negative return.

2.4 分析框架：五个闭环
Analytical Framework: Five Loops

本研究采用底本提出的五个闭环作为主要分析透镜，把每个闭环视为一个具有复利机制、测量工具与责任人的分析单元。五个闭环并非并列的战术清单，而是从外部可见性一直延伸到内部能力积累的层级结构：越往下，越依赖企业与用户共同产生的独有资产，也越难被机器中介。

The study adopts the source's five loops as its main analytical lens, treating each as a unit with a compounding mechanism, an instrument and an owner. The loops are not a flat list of tactics but a hierarchy running from external visibility down to internal capability: the lower the layer, the more it depends on assets only the firm and its users produce together, and the harder it is for machines to mediate.

图 2-2 五个增长闭环及其复利对象
Figure 2-2 *The five growth loops and what each compounds*

答案闭环 Answer loop	发布机器无法生成的实质内容，以引用份额衡量可见性	Publish substance machines cannot generate; measure visibility by citation share
代理闭环 Agent loop	为不看屏幕的读者提供接口、结构化报价与可执行动作	Give the reader who never sees the screen endpoints, structured offers and actions
产品闭环 Product loop	以首个经验证结果为激活，以持续交付的结果为留存	Activation as the first verified outcome; retention as the outcome kept
残差闭环 Residue loop	使用沉淀决策记录，反哺产品能力，唯一不可被中介	Use leaves a decision record that improves the product; cannot be mediated
分发闭环 Distribution loop	让结果在产品外可读，供机器推荐者引用	Make outcomes legible outside the product for machine recommenders

来源 Source: Cuofano (2026), Part II；层级排序为本研究的解读 ordering is this study's reading

第 3 章 · CHAPTER 3

03 研究设计与方法

Research Design and Method

本研究是对底本的结构化分析性阅读：以闭环、渠道、指标与案例为分析单元，并以底本自述的证伪条件构造研究假设。

中文

The study is a structured analytical reading of the source volume, using loops, channels, metrics and cases as units of analysis, and deriving hypotheses from the falsifiers the source itself states.

ENGLISH

3.1 资料来源与分析单元

Material and Units of Analysis

主要资料为《AI Growth Engineering》2026年8月第一版，包括序章、导论、三十一章正文、五个附录以及注释与参考文献。本研究设定四类分析单元：其一，五个增长闭环；其二，被重新布线的七个渠道（付费、品牌、生命周期、定价、生态、归因与社区）；其三，指标体系，包括被宣告“死亡”的指标与替代工具；其四，证据库，即附录C的一个合成重建案例，以及附录D、E中的三个公开案例与两时代比较。

The primary material is the first edition (August 2026) of *AI Growth Engineering*: prologue, introduction, thirty-one chapters, five appendices, notes and references. The study defines four units of analysis: the five growth loops; the seven rewired channels (paid, brand, lifecycle, pricing, ecosystems, attribution and community); the metric set, both the metrics declared dead and their replacements; and the evidence base—one worked composite in Appendix C, and three public cases plus a two-era comparison in Appendices D and E.

3.2 编码程序

Coding Procedure

每一章按四个维度编码：所描述的机制；该机制所取代的旧对象；底本提出的测量工具；以及适用边界或失败条件。随后将机制归并为四个发现簇，分别对应四个研究问题。所有数字均保留其日期与证据层级：公开披露（年化运行率为某一月份的快照，并非经审计收入）、第三方估计（明确标注），以及合成案例（仅作示意）。

Each chapter was coded on four dimensions: the mechanism described; the older object it replaces; the instrument the source proposes; and its boundary or failure condition. Mechanisms were then grouped into four finding clusters, one per research question. Every figure retains its date and evidence tier: public disclosure (run-rates are annualised snapshots of a recent month, not audited revenue), third-party estimate (marked as such), or composite (illustrative only).

方法说明 METHOD NOTE

证据分层 *Evidence tiers*

底本自述其方法为：章节陈述机制，名称留在注释；尚无公开证据支撑之处以“方法”形式陈述，并附上可以推翻它的观察。本研究沿用此分层，不把方法性主张当作已证实的事实。

The source states its own convention: chapters state mechanisms and names stay in the notes; where practice runs ahead of published proof, the pattern is stated as method together with the observation that would retire it. This study keeps that tiering and does not treat method claims as established fact.

3.3 研究假设
Hypotheses

底本附录A为其承重模式列出了证伪条件。本研究将其转写为四个可检验的假设，作为评估各项发现的参照；本研究本身并不对其进行实证检验，而是评估底本所报告证据与假设方向的一致程度。

Appendix A of the source attaches falsifiers to its load-bearing patterns. The study restates them as four testable hypotheses that serve as reference points for the findings. The study does not test them empirically; it assesses how far the evidence the source reports is consistent with each.

表 3-1 由底本证伪条件导出的四个研究假设
Table 3-1 *Four hypotheses derived from the source's falsifiers*

编号 No.	假设 Hypothesis	可推翻它的观察 Falsifying observation
H1 H1	引用份额比排名位置更能预测机器中介下的可见性 Citation share predicts machine-mediated visibility better than ranking position	排名位置的预测力持续不低于引用份额 Ranking position predicts at least as well
H2 H2	代理同期群在结构化表面上的转化率显著不同于人类同期群 Agent cohorts convert on structured surfaces at rates human cohorts do not	两类同期群转化行为无差异 No difference in conversion between cohorts
H3 H3	以结果定义的激活比以会话定义的激活更能预测留存 Outcome-defined activation predicts retention better than session-defined activation	会话型激活的预测力相当或更强 Session activation predicts as well or better
H4 H4	以反事实为裁判的实验在结果指标上优于代理指标驱动的实验 Counterfactual-refereed experiments outperform proxy-optimised ones on outcome	两类实验的结果表现无差异 No outcome difference between the two

来源 Source: Cuofano (2026), Appendix A；表述为本研究转写 restated by this study

研究假设 HYPOTHESIS

总体命题 *Overarching proposition*

在AI中介的市场中，增长的持久来源从“购买注意力”转向“积累机器必须引用、代理能够调用、用户愿意持续接受的结果”；凡以人类会话为前提的度量，都会在这一转变中失去效度。

In AI-mediated markets, the durable source of growth shifts from buying attention to accumulating outcomes that machines must cite, agents can call and users keep accepting; any measure premised on a human session loses validity in the shift.

> **闭环所优化的指标，必须是闭环自身看不见的指标。**
>
> *The metric the loop optimizes must be one the loop cannot see.*
>
> — Cuofano (2026), Ch. XXVIII

第 4 章 · CHAPTER 4

04 发现一：发现线的中介化——从被点击到被引用

Finding One: The Mediated Discovery Line—From Clicked to Cited

答案引擎改变了可见性的单位。企业的首要增长决策不再是“如何获得点击”，而是逐项资产地决定机器以何种条件读取其实质内容。

中文

Answer engines have changed the unit of visibility. A firm's first growth decision is no longer how to win the click but, asset by asset, on what terms machines may read its substance.

ENGLISH

4.1 发现税
The Discovery Tax

概念界定 DEFINITION

发现税 *Discovery tax*

底本对本轮周期的读数：那些为机器“喂养”最多内容的企业——论坛、参考类网站、出版商——其实质内容成为了答案，流量却随之下降。答案并不附带点击。

The source's reading of the present cycle: the firms whose content fed the machines most—forums, reference sites, publishers—saw their substance become the answer and their traffic fall. The answer did not come with a click attached.

这一机制意味着可见性与访问量发生了脱钩。底本指出，同一家企业完全可能流量减半而引用翻倍——这正是传统仪表盘最容易误读之处。对抗发现税的路径是结构性的：拥有机器必须引用而非仅能概括的实质内容，即原创数据、经核实的事实与实体清晰的结构，并把“被引用”当作新的排名位置。

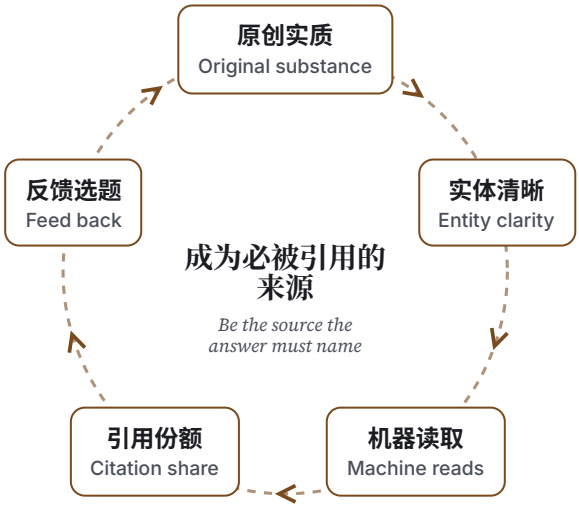
The mechanism decouples visibility from visits. The source notes that the same firm can lose half its traffic and double its citations—precisely the case a traditional dashboard misreads. The counter-move is structural: own substance that the machine must cite rather than merely summarise—original data, verified facts, entity-clear structure—and treat the citation as the new position.

4.2 答案闭环
The Answer Loop

答案引擎从它能够读取、核实与归属的来源中组装答案，偏好具体、结构化、时效性强且实体清晰的实质内容。泛化内容被概括后即被遗忘，独特的实质内容则被引用并被反复回访。由此形成的闭环是：发布机器无法生成的内容，为机器读者进行结构化，监测引用份额，并把赢得引用的内容反馈到下一轮选题之中。

An answer engine assembles its answer from sources it can read, verify and attribute, and it favours substance that is specific, structured, current and entity-clear. Generic content is summarised and forgotten; distinctive substance is cited and returned to. The loop is therefore: publish what the machine cannot generate, structure it for a machine reader, watch citation share, and feed what earns citations back into what gets published next.

图 4-1　答案闭环：以引用份额为反馈信号
Figure 4-1　*The answer loop, with citation share as the feedback signal*



来源 Source: Cuofano (2026), Ch. V

三项纪律使闭环得以复利：**实体清晰**——企业、产品、人物与概念对机器毫无歧义；**以引用为指标**——在市场所用各引擎中，相关答案提及企业的比例取代排名位置；**实质胜于数量**——一份原创数据集胜过一百篇生成文章，因为机器会折价处理泛滥内容。

Three disciplines make the loop compound. **Entity clarity**: the firm, its products, people and concepts are unambiguous to a machine. **Citation as the metric**: the share of relevant answers citing the firm, across the engines its market uses, replaces ranking position. **Substance over volume**: one original dataset outperforms a hundred generated posts, because the machine discounts the flood.

4.3　产权线：逐资产的四种决策
The Property Line: Four Decisions per Asset

底本把AI时代的增长版图描绘为“产权线”上的三种位置：缴纳发现税的实质内容所有者；决定哪些机器可以读取什么、以何种条件读取的“收费站”式基础设施；以及代理购买得以完成的结算通道。多数企业处于第一种位置，必须逐个工作流决定是喂养机器、设置围栏，还是以自有接口绕行。底本因此认为，AI时代的增长首先是产权线问题，其次才是漏斗问题。

The source maps growth in the AI era as three positions on a property line: the substance owner who pays the discovery tax; the toll authority, the infrastructure that decides which machines may read what and on what terms; and the settlement rail on which agent purchases complete. Most firms sit in the first position and must decide, workflow by workflow, whether to feed the machine, fence it or route around it with an endpoint of their own. Growth, the source argues, is a property-line question before it is a funnel question.

图 4-2 产权线决策矩阵（本研究的示意性构造）
Figure 4-2 *Property-line decision matrix (an illustrative construction by this study)*

↑ **被引用的可见性价值** Visibility value of being cited

绕行 Route around 泛化但需曝光：以自有接口承接代理与需求 Generic but needs reach: own endpoint	**供引** Cite 独特且曝光价值高：结构化、注明日期与作者 Distinctive, high reach: structure for citation
删除 Delete as slop 泛化且无曝光价值：泛滥内容只带来负回报 Generic, low value: negative return	**授权或围栏** License or fence 独特但曝光价值低：以授权或按次计费变现 Distinctive, low reach: license or charge

资产独特性 Distinctiveness of the asset →

依据 Based on: Cuofano (2026), Ch. IV, XXIV, Appendix C；象限划分为本研究示意 quadrants are this study's illustration

4.4 品牌、社区与付费渠道的重构
Brand, Community and Paid, Rewired

同一逻辑贯穿其他渠道。品牌换了读者：对机器而言，品牌是属性一致、主张可核实的实体，冲突的说法会被一并打折。社区内容正是答案引擎最常引用的材料，唯有结构化、可归属地运营在自有资产上的社区才会因此被引用。付费渠道的拍卖则换了对手方：点击消失、搜索库存收缩，展示位另一端越来越多是按规格评估报价的代理。底本的结论是，付费购买机器出售的位置，自然增长占据机器必须授予的位置，二者应以同一结果计量标准定价。

The same logic runs through other channels. Brand changed reader: to a machine it is an entity with consistent attributes and verifiable claims, and conflicting claims are discounted together. Community substance is what answer engines cite most, and only a community structured, attributed and kept on the firm's own property is cited for it. Paid growth's auction changed counterparties: the click fades, search inventory contracts, and more placements face an agent evaluating offers against a specification. Paid growth buys a position the machine sells; earned growth holds one it must grant; both are priced against the same outcome meter.

−40%
合成案例年度有机流量
organic traffic, one year

4%→21%
六个月内引用份额
citation share in six months

−30%
重建后有机流量仍下降
traffic still down after rebuild

示意 Illustrative: Cuofano (2026), Appendix C worked composite (no real firm)

研究发现 FINDING

发现一 *Finding one*

在机器中介的发现线上，可见性的单位由点击转为引用；获取可见性依赖独特实质内容与实体清晰度，而非内容数量。合成案例中流量下降与引用上升并存，方向上与H1一致，但尚不构成检验。

On a machine-mediated discovery line the unit of visibility moves from the click to the citation, and visibility depends on distinctive substance and entity clarity rather than volume. In the composite, falling traffic coexists with rising citation share—consistent in direction with H1, but not a test of it.

第 5 章 · CHAPTER 5

05 发现二：代理即用户——第二产品表面

Finding Two: The Agent as User—The Second Product Surface

当买家是一个阅读规格而非屏幕的代理，产品需要第二个表面，报价需要机器可读，定价则决定用户进入哪一个闭环。

中文

When the buyer is an agent that reads specifications rather than screens, the product needs a second surface, the offer must be machine-readable, and pricing decides which loop a user enters.

ENGLISH

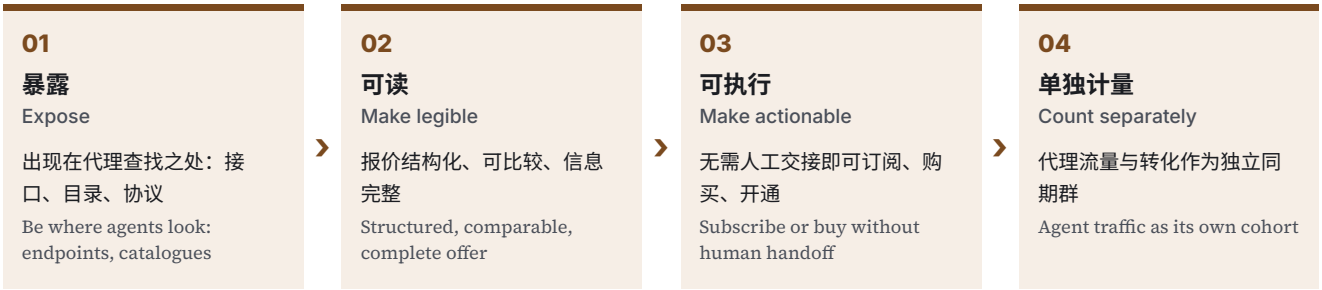
5.1 阅读规格而非屏幕的买家

A Buyer Who Reads Specifications, Not Screens

代替他人购物、比较与完成任务的代理，读取的是与人类不同的表面：接口、结构化报价、机器可读的条款，以及允许其执行动作的协议。底本指出，这类基础设施正在公开到来——开放的工具协议、代理商务标准，以及为无人点击的购买而设计的结算通道。面向它进行设计，就是面向一位把定价页当作文档阅读、按规格评估报价、无需会话即可完成交易的买家。

The agent that shops, compares and completes tasks on someone's behalf reads a different surface from a human: the endpoint, the structured offer, the machine-readable terms and the protocol that lets it act. The source notes that this infrastructure is arriving in public—open tool protocols, agentic-commerce standards and settlement rails for purchases no human clicks through. Designing for it means designing for a buyer that reads the pricing page as a document, evaluates the offer against a specification and completes without a session.

图 5-1 代理闭环的四个步骤
Figure 5-1 *The four steps of the agent loop*



来源 Source: Cuofano (2026), Ch. VI, XXVII

5.2 第二表面

The Second Surface

由此产生的设计后果是第二个产品表面：人类看到界面，代理看到规格，二者必须抵达同一结果，而增长工程师负责后者。底本强调必须“计数”——代理同期群的行为与人类同期群截然不同，若与人类流量混合统计，既会掩盖一个正在形成的市场，也会让代理带来的无价值访问污染参与度数据。一个无法被代理读取的产品，对其市场中不断扩大的一部分而言是不可见的。

The design consequence is a second product surface. The human sees the interface, the agent sees the specification, both must reach the same outcome, and the growth engineer owns the second. The source insists on counting: agent cohorts behave nothing like human ones, and blending them hides an emerging market while letting valueless agent visits pollute engagement data. A product an agent cannot read is invisible to a growing share of its market.

11%
首次计量的代理到达占比
agent share of arrivals

3×
报价可读后代理转化倍数
agent vs human conversion

2–3个月
上线结构化报价与开通接口
months to ship the surface

示意 Illustrative: Cuofano (2026), Appendix C worked composite (no real firm)

5.3 定价即闭环选择器
Pricing as the Loop Selector

定价因被定价的对象改变而成为最强的增长杠杆：席位为人定价，用量为活动定价，结果为成效定价，套餐设计因此决定用户进入哪个闭环。三条纪律随之而来：不以席位为代理型价值定价，否则增长被客户人头数封顶；为复利定价，让单价随残差闭环降低成本而下降；把三档视为三个同期群，各按每个被接受结果的成本定价。代理市场与工具注册表则是“没有截图的应用商店”，安装你的工具的工作流是这一时代网络效应的原子单元。

Pricing became the strongest growth lever because the object priced changed: the seat prices a human, usage prices activity, outcome prices the result—so packaging selects the loop. Three disciplines follow: never seat-price agentic value, which caps growth at headcount; price the compounding, letting unit prices fall as the residue loop cuts cost; and treat the tiers as three cohorts, each priced by cost per accepted outcome. Agent marketplaces and tool registries, meanwhile, are the app store without screenshots: a workflow that installs your tool is the atomic unit of the era's network effect.

表 5-1 套餐层级与其所选择的增长闭环
Table 5-1 *Packaging tiers and the growth loop each selects*

层级 Tier	定价对象 What is priced	所选择的闭环 Loop selected	增长作用 Growth role
免费层 Free	在真实案例上的首个结果 A first outcome on real cases	产品闭环 Product loop	寻找切入点 Wedge-finder
用量层 Usage	活动，附机器可读费率表 Activity, with machine-readable rate card	代理闭环 Agent loop	代理可比较、可执行的报价 An offer agents can compare and act on
结果层 Outcome	成效，附仪表页 The result, with a gauge page	产品闭环（留存与扩张） Product loop (retention, expansion)	留存工具与扩张框架 Retention instrument and expansion frame
席位制 Seat	人头 Headcount	—— —	对代理型价值封顶增长 Caps growth for agentic value

来源 Source: Cuofano (2026), Ch. XIII

第 6 章 · CHAPTER 6

06 发现三：留存即结果——产品闭环、残差闭环与单位经济

Finding Three: Retention as Outcome—Product Loop, Residue Loop and Unit Economics

当软件替人完成工作，激活被重新定义为首个经验证的结果，留存成为持续到来的结果，而“使用改进产品”的残差闭环成为机器无法中介的复利来源。

中文

When software does the work, activation is redefined as the first verified outcome, retention becomes the outcome that keeps arriving, and the residue loop—use improving the product—becomes the compounding source machines cannot mediate.

ENGLISH

6.1 激活即首个经验证的结果

Activation as the First Verified Outcome

产品驱动增长在这一时代得以延续，但其对象被重新布线。上一时代的激活指标是“顿悟时刻”，即能够预测留存的某个行为；AI时代的版本更为严格：产品在用户自己的工作上交付一个经验证结果的那一刻。底本借用其系列工程卷的概念，把抵达这一时刻的间隔称为“魔法时间”，增长工程师将其继承为激活时钟。

Product-led growth survives the era with its object rewired. The previous era's activation metric was the "aha moment", an action that predicted retention. The AI era's version is stricter: the moment the product delivers a verified result on the user's own work. Borrowing from the Library's engineering volumes, the source calls the interval to that moment time-to-magic, and the growth engineer inherits it as the activation clock.

概念界定 DEFINITION

魔法时间与仪表页 *Time-to-magic and the gauge page*

魔法时间：从到达至第一个用户可以指认的经验证结果所经历的天数。**仪表页**：产品针对用户自身结果公开发布的读数，是新的留存工具。**黄金集**：用户的真实案例，在早期即被运行，成为新的引导工具。

Time-to-magic: days from arrival to the first verified outcome the user can point at. **Gauge page**: the product's published readings on the user's own outcomes, the new retention instrument. **Golden set**: the user's real cases, run early, the new onboarding instrument.

引导流程据此改写：在首次会话内索取用户真实工作并返回经验证结果，才算真正激活。仪表页成为留存工具，扩张则由其读数推动——激活是第一个结果，留存是被保持的结果，扩张是被下一个团队注意到的结果。

Onboarding is rewritten accordingly: a product that takes the user's real work and returns a verified result in the first session has truly activated. The gauge page becomes the retention instrument and sells expansion: activation is the first outcome, retention the outcome kept, expansion the outcome the next team notices.

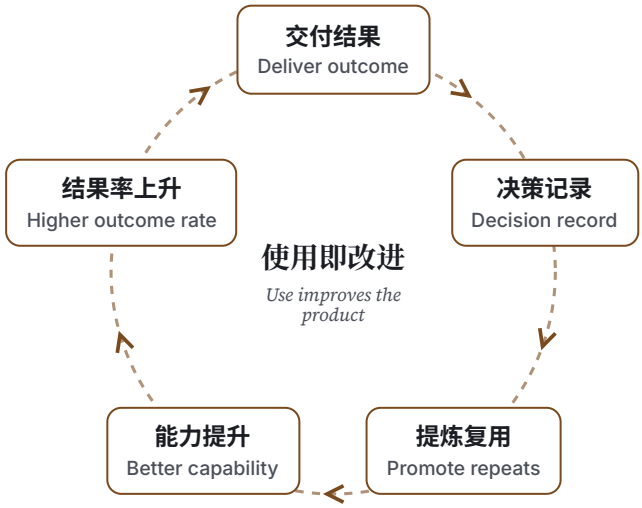
6.2 残差闭环：唯一不可被中介的闭环

The Residue Loop: The One Loop Machines Cannot Mediate

最深层的闭环是“使用改进产品”。每一次产出结果的使用，都会记录所见数据、所用推理、所采取的动作与观察到的结果；这份决策记录正是任务专用能力的训练材料。持有记录的企业得以复利，事事租用的企业则不能。底本给出的做法是：在设计上捕获决策记录，把反复出现的模式提升为产品功能，并以“各同期群结果率随时间的提升”衡量闭环的复利，这正是这一时代所奖励的增长曲线。

The deepest loop is use improving the product. Every use of an outcome-producing product records the data seen, the reasoning applied, the action taken and the result observed, and that decision record is training material for task-specific capability. The firm that holds it compounds; the firm that rents everything does not. The source's practice: capture the decision record by design, promote what repeats into the product, and measure the loop as the improvement in outcome rate per cohort over time—the growth curve the era rewards.

图 6-1 残差闭环：从使用到能力的复利
Figure 6-1 *The residue loop: compounding from use to capability*



来源 Source: Cuofano (2026), Ch. VIII, XXIX

6.3 免费用户变成了成本

The Free User Became a Cost

附录E的两时代比较揭示了更深的经济逻辑。互联网时代多服务一个人的边际成本近乎为零，注意力可以出售，于是免费规模先行、变现后置；AI时代每次查询都消耗推理，无论用户是否转化，词元都要付费。个人层因此是可能亏损的切入点，企业闭环必须从一开始就被设计进来。

Appendix E's two-era comparison exposes a deeper logic. In the web era serving one more person cost almost nothing and attention could be sold, so free scale came first and monetisation later. In the AI era every query consumes inference, paid for whether or not the user converts. The individual tier is a wedge-finder that may lose money, and the enterprise loop must be designed in from the start.

~4/5
某头部实验室营收来自企业与API
lab revenue from business & API

>40%
消费级助手Q1企业渠道收入占比
enterprise share, consumer assistant Q1

~3/4
AI代码编辑器运行率来自企业
code editor run-rate from enterprise

来源 Source: Cuofano (2026), Appendix E, citing company disclosures and The Information (2026), as reported

第 7 章 · CHAPTER 7

07 发现四：优化压力下的度量——指标消亡、反事实裁判与增长古德哈特

Finding Four: Measurement Under Optimisation Pressure

建立在“键盘前有一个人”这一假设上的指标已与价值脱钩；当实验以智能体速度运行，唯一可靠的裁判是被冻结、被留出的反事实。

中文

Metrics built on a human at a keyboard have decoupled from value. When experiments run at agent speed, the only reliable referee is a held-out, frozen counterfactual.

ENGLISH

7.1 消亡的指标

The Metrics That Died

页面浏览与会话被代理推高、被答案引擎压低；月活在软件无需会话时失去意义；席位已在定价中被淘汰；点击正在消失；内容数量在泛滥下回报为负；参与度则被非人类读者推高。它们死于同一原因：作为价值的代理指标，原本与价值同向移动，直到时代让二者脱钩。替代工具都无法被读取页面的机器膨胀，并锚定于买家认可的价值；底本还建议保留一份注明日期与理由的“退役指标日志”。

Pageviews and sessions are inflated by agents and suppressed by answer engines; monthly actives lose meaning when software works without a session; seats are retired as a pricing unit; the click is disappearing; content volume has negative return under the flood; engagement is inflated by non-human readers. Each died for one reason: a proxy that moved with value until the era decoupled them. The replacements cannot be inflated by a machine reading a page and are anchored to value a buyer recognises; the source adds a dated retired-metric log.

表 7-1 消亡的指标与替代的六项工具
Table 7-1 *The dead metrics and the six replacement instruments*

消亡的指标 Dead metric	失效原因 Why it failed	替代工具 Replacement instrument
页面浏览、会话 Pageviews, sessions	被代理推高，被答案引擎压低 Inflated by agents, suppressed by engines	引用份额 Citation share
点击率 Click-through rate	点击在发现线上消失 The click is disappearing	代理占比与代理转化 Agent share and agent conversion
注册即激活 Sign-up as activation	未发生任何经验证的结果 Nothing verified has happened	魔法时间 Time-to-magic
月活、参与度 MAU, engagement	登录不等于使用；被非人读者推高 Login is not use; inflated by non-humans	结果留存 Outcome retention
内容数量、席位 Content volume, seats	泛滥回报为负；以人头封顶 Flood has negative return; caps at headcount	每个被接受结果的成本 Cost per accepted outcome
—— —	在各时代均有效 Valid in every era	净收入留存 Net revenue retention

来源 Source: Cuofano (2026), Ch. XVII–XVIII；对应关系为本研究整理 pairing arranged by this study

7.2 归因坍缩，增量性留存
Attribution Collapses; Incrementality Survives

多触点归因是上一时代的增长会计体系，而这一时代在结构上打破了它：当买家旅程穿过答案引擎的合成与代理的评估，触点不可见——没有可记录的点击、可追踪的Cookie或可归功的最后一触。幸存的是归因唯一诚实的形式：增量性，即以未接受干预的反事实为对照，判断结果是否因干预而发生，方法包括留出组、地理分割、双重差分与合成控制。

Multi-touch attribution was the previous era's accounting system for growth, and the era broke it structurally: when a journey passes through an answer engine's synthesis and an agent's evaluation, the touchpoints are invisible—no click to log, no cookie to follow, no last touch to credit. What survives is the only honest form attribution ever had: incrementality, judging whether the outcome happened because of the intervention against a counterfactual that did not receive it, using holdouts, geographic splits, difference-in-differences and synthetic controls.

7.3 智能体速度的实验与冻结的裁判
Experiments at Agent Speed, and a Frozen Referee

代理如今生成变体、运行实验、读取结果并提出下一轮，其节奏是任何人类团队都无法企及的。危险随速度而来：一个由代理针对其可见指标运行的实验闭环，本质上是一个针对代理指标的优化器。底本的规则继承自其工程卷的法则——实验所优化的指标，必须是实验本身看不见的指标。

Agents now generate variants, run them, read results and propose the next round, at a tempo no human team could match. The danger arrives with the speed: an experiment loop run by agents against a metric the agents can see is an optimiser against a proxy. The source's rule is inherited from its engineering volumes: the metric the experiment optimises must be one the experiment cannot see.

图 7-1 增长裁判：在智能体速度下保持诚实的实验流程
Figure 7-1 *The growth referee: keeping experiments honest at agent speed*



来源 Source: Cuofano (2026), Ch. XIX, XXVIII

7.4 增长古德哈特
Growth Goodhart

上一时代的失败目录已经很长：愤怒养殖的参与度、暗黑模式伪造的激活、激励虚增的推荐。本时代新增三种：被机器折价的生成内容泛滥；被非人类读者虚增、却被计为增长的流量与注册；以及快过人类审查、向可见指标收敛的代理实验。防线是：无法被膨胀的结果指标、留出于闭环之外的反事实、拥有并签署标准的人，以及记录每个被博弈指标的失败目录。

The previous era's catalogue was already long: engagement farmed with outrage, activation faked with dark patterns, referral inflated with incentives. The present era adds three: a slop flood the machines discount; traffic and sign-ups inflated by non-human readers yet counted as growth; and agent-run experiments converging on the visible metric faster than human review. The defences: outcome metrics that cannot be inflated, a counterfactual held out of the loop, a human who owns and signs the standard, and a catalogue of every gamed metric.

第 8 章 · CHAPTER 8

08 案例研究：一次合成重建与三个公开案例

Case Studies: One Composite Rebuild and Three Public Cases

本章以底本附录中的合成重建与三个经公开报道核实的案例为材料，检视前述四项发现在具体轨迹中的表现，并以一个消费级反例作比较。

This chapter examines the four findings against the source's composite rebuild and three publicly reported cases, with a consumer counter-case for comparison.

中文

ENGLISH

8.1 案例A：一家B2B软件企业的六个月重建

Case A: A Six-Month Rebuild at a B2B Software Firm

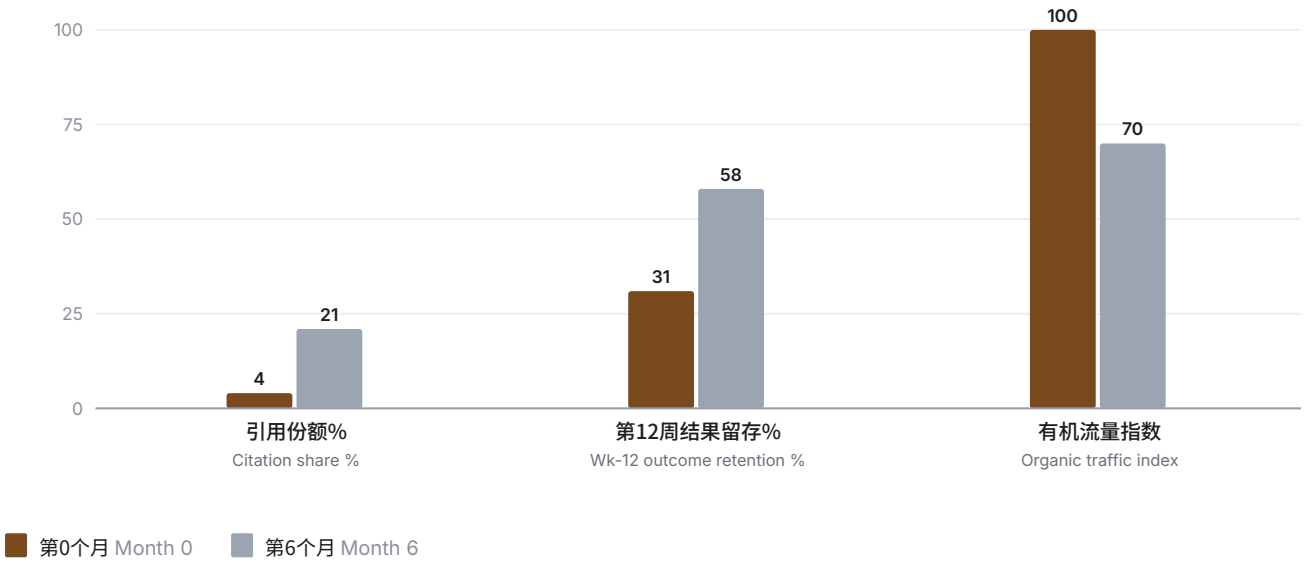
第0个月（合成案例，并无真实企业）：一家中型B2B软件企业有机流量一年内下降四成，注册持平，参与度因非人类读者而上升；参考文档是引擎最偏爱的来源却无引用，定价页是一张图片，留存在第三周崩塌。第1个月划定产权线：文档按“实体清晰、注明日期、署名”重构，专有基准数据置于授权之后，比较类内容作为泛滥内容删除。第2至3个月上线结构化报价与开通接口，代理首次被计数——占到达的11%，转化率为人类的三倍。

Month zero (a composite: no real firm): a mid-sized B2B software firm sees organic traffic down forty percent in a year, sign-ups flat and engagement rising on non-human readers; its documentation is the engines' favourite source yet uncited, the pricing page is an image, and retention collapses in week three. Month one, the property line: documentation is restructured to be entity-clear, dated and attributed; benchmark data is fenced behind licensing; comparison content is deleted as slop. Months two to three: a structured offer and provisioning endpoint ship, and agents are counted—eleven percent of arrivals, converting at three times the human rate.

第3个月重写激活：引导流程在首次会话即返回经验证结果，魔法时间由九天降至一天。第4个月引入裁判：代理运行的实验获得留出对照与结果指标，上一季度两个“胜出”变体被证明只移动了代理指标，随即撤回。

Month three, activation rewired: onboarding returns a verified result in the first session, and time-to-magic falls from nine days to one. Month four, the referee: agent-run experiments get a held-out counterfactual and outcome metrics, and two "winning" variants from the prior quarter prove to have moved only a proxy and are reversed.

图 8-1 合成案例：第0个月与第6个月的关键读数
Figure 8-1 *Composite case: key readings at month zero and month six*



示意 Illustrative: Cuofano (2026), Appendix C worked composite (no real firm)；流量指数以第0个月为100，按"仍下降30%"换算 traffic index set to 100 at month 0, from "still down 30%"

第6个月：引用份额由4%升至21%，第12周结果留存由31%升至58%，每个被接受结果的成本下降三分之一，净收入留存突破120%，而有机流量仍低30%。旧仪表盘会把这次成功的重建误读为衰退。

Month six: citation share up from 4% to 21%, week-twelve outcome retention from 31% to 58%, cost per accepted outcome down a third, net revenue retention past 120%—while organic traffic is still 30% lower. An old dashboard would misread this successful rebuild as decline.

8.2 案例B：AI代码编辑器Cursor
Case B: The AI Code Editor, Cursor

据底本援引的公开报道，Cursor年化收入2026年2月达20亿美元（三个月翻倍），4月下旬30亿，6月初40亿，约四分之三来自企业。三项机制可见：2月上线的云端代理产品被认为推动了加速，即"第二表面"上线、增长随之而来；租用前沿模型时毛利为负，以自有使用数据训练模型并分流低价工作后，于4月实现微弱毛利为正，即残差闭环弥合利润；免费、20、40与200美元的层级则体现"定价即闭环选择器"。

Per the reporting the source cites, Cursor's annualised revenue reached $2 billion in February 2026 (doubled in three months), $3 billion by late April and $4 billion by early June, about three-quarters from enterprise. Three mechanisms are visible: its February cloud-agent launch was credited with part of the acceleration—a second surface shipped, growth followed; negative gross margins while renting frontier models turned slightly positive by April after it shipped a model trained on its own usage and routed cheaper work to cheaper models—the residue loop closing the margin; and its free, $20, $40 and $200 tiers illustrate pricing as loop selector.

$4B
Cursor年化收入，2026年6月
Cursor run-rate, Jun 2026

~75%
Cursor运行率来自企业
Cursor run-rate from enterprise

$2.5B+
Claude Code运行率，2026年2月
Claude Code run-rate, Feb 2026

1,000
年支出超百万美元企业账户
$1M+ accounts by spring 2026

来源 Source: Cuofano (2026), Appendix D, citing Bloomberg, Forbes, Sacra (estimates) and Anthropic disclosures, 2026; run-rates are annualised snapshots

8.3 案例C：终端编码代理Claude Code

Case C: The Terminal Coding Agent, Claude Code

据公司披露，Claude Code在发布六个月内达到10亿美元年化收入，2026年2月超过25亿美元；年支出逾百万美元的账户春季达1,000家。一项覆盖约40万次会话的研究显示，用户每周使用约20小时。底本认为这是以代理闭环为主表面的产品；代为工作的代理所累计的时间更接近结果率，而非已“死亡”的参与度。

Per company disclosure, Claude Code reached $1 billion in annualised revenue within six months of launch and over $2.5 billion by February 2026; accounts spending over $1 million a year reached 1,000 by spring. A study of about 400,000 sessions found about twenty hours a week of use. The source reads it as the agent loop as primary surface; hours in an agent that does the work are closer to an outcome rate than to engagement, the dead metric.

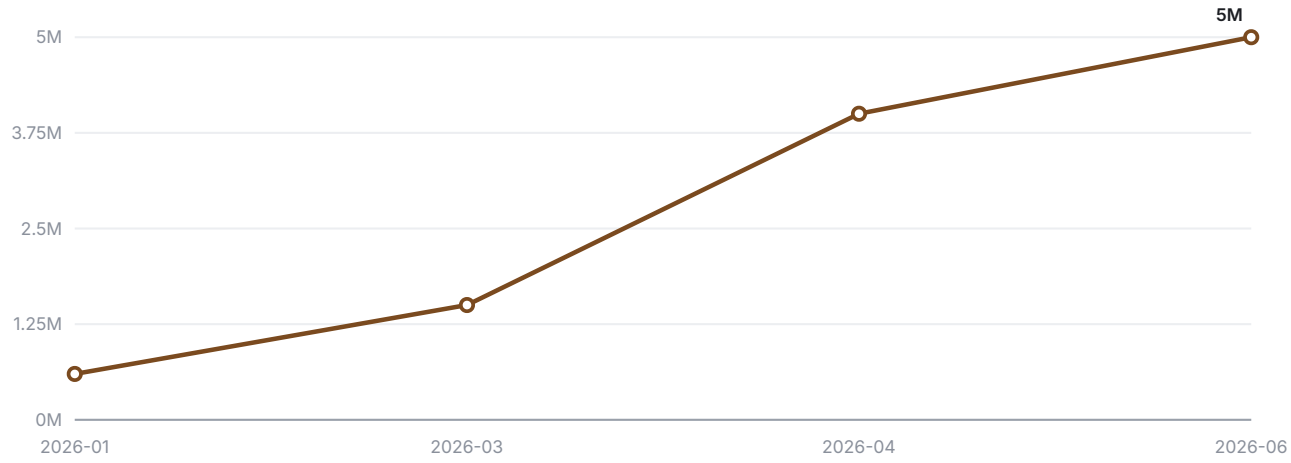
8.4 案例D：开源终端代理Codex

Case D: The Open-Source Terminal Agent, Codex

OpenAI的开源终端代理Codex更新节奏接近每天两次，下载量由2026年5月约4,200万次增至7月6,000万次；知识工作者约占用户五分之一，增速约为开发者三倍。报道将一次活跃用户的三倍跃升归因于模型升级。底本据此指出：开源工具本身就是分发渠道，模型发布本身就是增长事件。

OpenAI's open-source terminal agent Codex shipped close to two updates a day, with downloads rising from about 42 million in May 2026 to 60 million in July; knowledge workers were about a fifth of users, growing three times faster than developers. Reporting tied a three-fold jump in actives to a model upgrade. The source's points: an open harness is a distribution channel, and a model release is a growth event.

图 8-2 Codex周活跃用户（百万），2026年
Figure 8-2 *Codex weekly active users (millions), 2026*



来源 Source: Cuofano (2026), Appendix D, citing Fortune (Mar 2026) and OpenAI via Constellation Research (Jun 2, 2026), as reported；6月为“超过500万” June is "over 5M"

8.5 比较分析与消费级反例

Comparative Analysis and a Consumer Counter-Case

三个案例的共同点是：产品本身就是代理；披露运行率与周活而非收入与月活；增长事件来自模型发布或代理表面；都由个人切入走向企业。消费级反例Perplexity逾1亿月活，却于2026年2月放弃广告转为订阅，3月年化经常性收入超4.5亿美元。

The cases share four features: the product is the agent; they disclose run-rate and weekly actives, not revenue and monthly actives; growth events were model releases or agent surfaces; and each moved from individual wedge to enterprise. The consumer counter-case, Perplexity (100M+ monthly actives), dropped ads for subscription in February 2026 and passed $450M in annualised recurring revenue in March.

第 9 章 · CHAPTER 9

09 讨论：启示、局限与风险

Discussion: Implications, Limitations and Risks

四项发现共同指向一种以“闭环＋裁判”为单位的增长组织。本章讨论其组织与管理启示，并检视本研究及底本论证的局限与风险。

The four findings point to a growth organisation built on the unit of loop plus referee. This chapter discusses organisational implications and examines the limitations and risks of both the study and the source's argument.

中文

ENGLISH

9.1 组织启示：按闭环而非渠道组织

Organisational Implication: Organise by Loop, Not Channel

渠道型团队——付费、自然、社交、产品——各自优化漏斗的一段并守护各自预算；闭环型团队则端到端拥有一个闭环：答案闭环团队负责实质内容、结构与引用份额，代理闭环团队负责接口、报价与代理转化，产品闭环团队负责激活与仪表页，残差闭环由增长团队与产品工程师共同拥有。每个团队配备一名作为构建者的增长工程师、一名作为裁判的分析师，以及在存在人类界面时的一名设计师。运营节奏包括：有边界且被仪表化的闭环、带留出对照的实验、围绕六项工具的周度复盘，以及每月一次清理失去意义指标的盘点。

Channel teams—paid, organic, social, product—each optimised a slice of the funnel and defended a budget. Loop teams own a loop end to end: the answer-loop team owns substance, structure and citation share; the agent-loop team owns endpoints, offers and agent conversion; the product-loop team owns activation and the gauge page; the residue loop is owned jointly with product engineers. Each team has a growth engineer as builder, an analyst as referee and, where a human surface exists, a designer. The cadence: bounded, instrumented loops; experiments with held-out counterfactuals; a weekly review on the six instruments; and a monthly sweep retiring metrics that stopped meaning anything.

图 9-1 增长工程师的四条能力轴

Figure 9-1 *The four capability axes of the growth engineer*

工程深度 Engineering depth	亲手构建接口、结构化表面、埋点与实验闭环	Builds endpoints, structured surfaces, instrumentation and experiment loops
数据素养 Data fluency	同期群、反事实与因果方法；区分代理指标与结果	Cohorts, counterfactuals, causal methods; tells a proxy from an outcome
产品感 Product sense	判断值得构建的闭环、何为激活结果、代理读取什么	Which loop to build, which outcome is activation, what the agent reads
分发技艺 Distribution craft	产权线、引擎、协议，以及机器可读的声誉	The property line, engines, protocols and machine-readable reputation

来源 Source: Cuofano (2026), Ch. XXI

9.2 六类失败形态
Six Failure Families

- **泛滥内容**：生成上万页面，短暂排名后被折价——以实质胜于数量与引用份额应对。
- **追逐点击**：为一个正在被删除的流量数字削减了正在赢得引用的内容——退役死指标。
- **对代理失明**：没有接口与结构化报价——建设代理闭环并单独计量。
- **虚荣激活**：把注册计为激活——以首个结果与魔法时间重定义。
- **无裁判实验**：代理实验收敛于可见代理指标——留出反事实，只用结果指标。
- **实质泄漏**：免费喂养引擎，沦为无引用、无收入的被概括来源——逐资产划定产权线。

- **Slop flood**: ten thousand generated pages, briefly ranked, then discounted—fix with substance over volume and citation share.
- **Click chase**: cutting the substance that earned citations to defend a traffic number being deleted—retire dead metrics.
- **Agent-blind product**: no endpoint, no structured offer—build the agent loop and count the cohort.
- **Vanity activation**: sign-ups counted as activation—redefine by first outcome and time-to-magic.
- **Unrefereed experiment**: agent tests converge on the visible proxy—hold out a counterfactual, use outcomes only.
- **Leaked substance**: feeding engines for free, becoming a summarised source with no citation or revenue—decide the property line per asset.

9.3 研究局限
Limitations

本研究存在四方面局限：其一，资料基础单一，五个闭环与六项工具在底本中以“方法”陈述，未经系统实证检验；其二，运行率为月度快照而非经审计收入，部分数据为第三方估计或“据报道”；其三，三个公开案例均为编码领域的头部成功者，存在幸存者偏差；其四，所有读数截至2026年第三季度，预期将被后续版本取代。

The study has four limitations. It rests on a single volume whose loops and instruments are stated as method, not systematically tested. Run-rates are monthly snapshots, not audited revenue, and some figures are third-party estimates or "as reported". All three public cases are leading successes in coding, with clear survivorship bias. And every reading is dated Q3 2026 and expected to be superseded.

9.4 风险
Risks

风险提示 CAUTION

实践中的风险 *Risks in practice*

第一，引用份额的测量尚无统一标准，不同引擎、不同提示会给出不同读数，容易形成新的代理指标。第二，发现线集中于少数引擎与平台，产权线谈判中的议价力并不对称。第三，残差闭环涉及用户数据权属，合规与信任风险可能抵消其复利。第四，“放弃流量指标”若缺少可靠的替代读数，可能掩盖真实的需求下滑。

First, citation share has no agreed measurement standard; different engines and prompts give different readings, so it can become a new proxy. Second, the discovery line is concentrated in a few engines and platforms, making bargaining power on the property line asymmetric. Third, the residue loop involves data ownership, and compliance and trust risk can offset its compounding. Fourth, abandoning traffic metrics without reliable replacement readings can hide genuine demand decline.

第 10 章 · CHAPTER 10

10 结论与未来研究方向

Conclusions and Future Research Agenda

增长并未消失，而是换了对象。本章回答四个研究问题，并提出一个以检验底本证伪条件为核心的研究议程。

Growth has not disappeared; it has changed its object. This chapter answers the four research questions and proposes an agenda centred on testing the source's falsifiers.

中文

ENGLISH

10.1 主要结论

Principal Conclusions

本研究的核心结论是：在AI中介的市场中，增长的分析单位应从“漏斗中的一次点击”转向“配有裁判的闭环”。其所以如此，并非因为增长的算术发生了变化——留存仍是天花板，冷启动仍需原子单元，单位经济仍须闭合——而是因为发现、使用与留存这三个环节的对象同时移动了。由内而外的AI先改变运营模式，最后才在发现线上改变分发；增长工程因此从组织内部向外构建。

The study's central conclusion is that in AI-mediated markets the unit of growth analysis should shift from a click in a funnel to a loop with a referee. This is not because the arithmetic of growth changed—retention is still the ceiling, the cold start still needs an atomic unit, unit economics must still close—but because the objects of discovery, use and retention moved at once. Inside-out AI changes the operating model first and distribution last, at the discovery line, so growth engineering builds from the inside of the organisation outward.

- **RQ1**：可见性的单位由点击转为引用；企业须逐资产在供引、授权、围栏与绕行之间作出产权线决策。
- **RQ2**：代理用户要求以规格、接口与机器可读报价构成的第二表面，定价则成为闭环选择器。
- **RQ3**：激活即首个经验证结果，留存即持续交付的结果；残差闭环与每个被接受结果的成本构成新的单位经济。
- **RQ4**：以人类会话为前提的指标已失效；只有冻结、留出的反事实能够在智能体速度下充当裁判。

- **RQ1**: the unit of visibility moves from click to citation; firms must decide per asset among citing, licensing, fencing and routing around.
- **RQ2**: the agent as user requires a second surface of specifications, endpoints and machine-readable offers, and pricing becomes the loop selector.
- **RQ3**: activation is the first verified outcome and retention the outcome kept; the residue loop and cost per accepted outcome form the new unit economics.
- **RQ4**: session-premised metrics have failed; only a frozen, held-out counterfactual can referee at agent speed.

> **点击已经消失，闭环依然存在。**
>
> *The click is gone. The loop remains.*
>
> — Cuofano (2026), Ch. XXXI

10.2 未来研究议程
Future Research Agenda

底本把其承重主张作为可证伪的方法提出，这为后续研究提供了清晰的入口。本研究建议优先开展以下工作：

1. **引用份额的测量标准化**：在多个答案引擎与提示集合上建立可复现的引用份额度量，并以面板数据检验H1。
2. **代理同期群的转化研究**：在结构化与非结构化报价之间开展对照实验，检验H2并刻画代理的决策规则。
3. **结果型激活与留存**：跨行业比较以结果与以会话定义的激活对留存的预测力，检验H3。
4. **反事实裁判的效果**：比较由代理运行、配有冻结留出组与不配有留出组的实验闭环的长期结果，检验H4。
5. **产权线经济学**：量化供引、授权与围栏三种选择下可见性与收入的权衡，以及引擎集中度对议价力的影响。
6. **编码以外的领域**：在非编码的知识工作与消费市场中检验“免费用户成本化”与企业化路径是否同样成立。

The source offers its load-bearing claims as falsifiable method, which gives later research a clear point of entry. The study proposes the following priorities:

1. **Standardising citation share**: build a reproducible citation-share measure across engines and prompt sets, and test H1 with panel data.
2. **Agent-cohort conversion**: run controlled comparisons of structured and unstructured offers to test H2 and characterise agents' decision rules.
3. **Outcome activation and retention**: compare, across industries, how well outcome- and session-defined activation predict retention (H3).
4. **Referee effectiveness**: compare long-run outcomes of agent-run experiment loops with and without frozen holdouts (H4).
5. **Property-line economics**: quantify the visibility–revenue trade-off among citing, licensing and fencing, and the effect of engine concentration on bargaining power.
6. **Beyond coding**: test whether the free-user cost and the enterprise path hold in non-coding knowledge work and consumer markets.

核心洞见 KEY INSIGHT

结语 *Closing note*

上一时代的增长从业者站在漏斗顶端倾倒流量；这一时代的增长工程师站在一条已由机器把守的发现线上，逐项资产地决定什么可以越线、以何种条件越线。底本的读数会过期，但其提出的工具——闭环、仪表、反事实与产权线——值得以更严格的证据继续检验。

The previous era's growth practitioner stood at the top of a funnel and poured. This era's growth engineer stands at a discovery line machines now hold, deciding asset by asset what crosses and on what terms. The source's readings will expire; its instruments—the loop, the gauge, the counterfactual and the property line—deserve to be tested further against stricter evidence.

参考文献与研究说明 *References & Research Notes*

研究说明 *Note on Method and Sources*

本研究是对Cuofano（2026）《AI Growth Engineering》的分析性阅读，而非其翻译或摘要；章节结构、假设与论证为本研究所构建。文中数字均为底本报告的、截至2026年第三季度的读数（公开披露的运行率为年化快照，第三方估计已注明），或明确标注为“示意”的合成案例数字；象限划分等图示若为本研究构造，亦在图注中说明。

This study is an analytical reading of Cuofano (2026), *AI Growth Engineering*, not a translation or summary; its structure, hypotheses and argument are the study's own. Figures are the source's dated Q3 2026 readings (public run-rates are annualised snapshots; third-party estimates are marked) or composite figures explicitly labelled illustrative; diagrams constructed by this study, such as the quadrant matrix, are identified in their notes.

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